LECTURE: CLASSIFICATION

Map the o/p probabilities to discrete class labels

Assign label based on maximum probability

$\hat{y}_{new} = 1$ if $P(y_{new} = 1 \mid x_{new}, w) \geq 0.6$

Threshold's can be optimized using Cross Validation (CV) data dependent

$$y(x) = \sigma(w^T x) = \frac{1}{1 + e^{-w^T x}} = P(t=1 \mid x, w)$$

For binary label $t_i \in \{0, 1\}$

$P(t=0 \mid x, w) = 1 - P(t=1 \mid x, w)$

Individual Class Likelihood

$P(t_i = 1 \mid x_i, w) = \sigma(w^T x_i)$

$P(t_i = 0 \mid x_i, w) = 1 - \sigma(w^T x_i)$

(Bernoulli Trick)

Combined Probability Expression:

$$P(t_i \mid x_i, w) = [\sigma(w^T x_i)]^{t_i} [1 - \sigma(w^T x_i)]^{1-t_i} = y_i^{t_i} (1 - y_i)^{1-t_i}$$

Log-likelihood function:

$$\log p(t_i \mid x_i, w) = t_i \log[\sigma(w^T x_i)] + (1 - t_i) \log[1 - \sigma(w^T x_i)]$$

Optimization Objective: optimal weights w^* by minimizing the negative log-likelihood (equivalent to maximizing log-likelihood)

$$w^* = \arg\min_w J(w)$$

Cost Function: $J(w)$ Negative log-likelihood loss is:

$$J(w) = -\sum_{i=1}^{N} \log p(t_i \mid x_i, w)$$

L2 Regularization: Add L2 penalty term $[\lambda(\|w\|^2)]$ to prevent overfitting. $J(w) = -\sum_{i=1}^{N} \log p(t_i \mid x_i, w) + \lambda \|w\|^2$

$$\frac{1}{2} \sum \{t_n - w^T \phi(x_n)\}^2 + \frac{\lambda}{2} \sum_{j=1}^{M} |w_j|^q \qquad E_w(w) \sim$$

Generalized Regularization

- classification: Nearest Neighbour

Regression task to classification task

Oranges vs Lemons

K-Nearest Neighbor

↳ Rely on more than 1 nearest point

choosing K → important task

K = too low not a good prediction

$$\|x_1 - x_{new}\|_2^2 = \sum_i (x_1(i) - x_{new}(i))^2$$

choice of distance metric — depends on data

Cross-validation

K=1

simple

inefficient

Vulnerability to class imbalance

Sensitivity to noise at small K

Logistic Regression:

Data Points: $(x_1, t_1), (x_2, t_2), \ldots, (x_N, t_N)$ $\quad x_i = [x_{i1}, x_{i2}]^T$

Target labels: $t_i \in \{-1, 1\}$

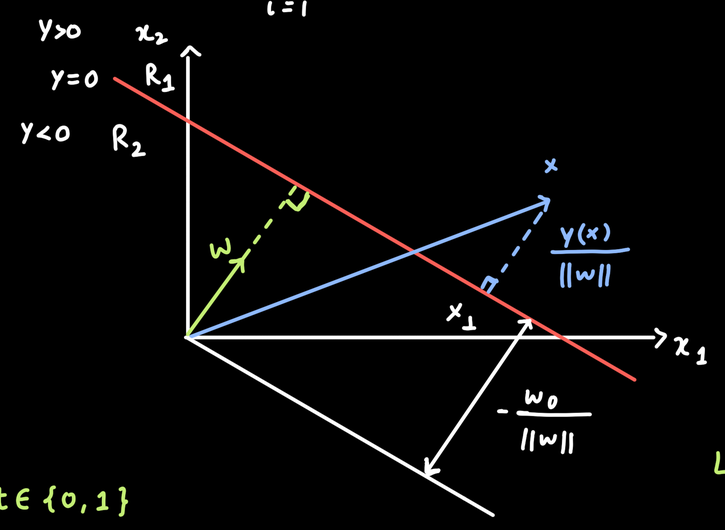
Decision Boundary & Linear Model

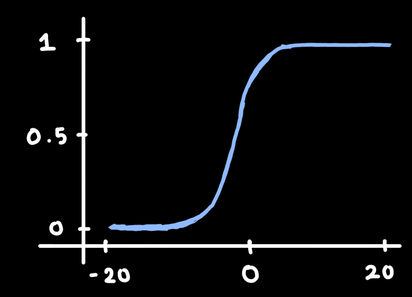
- classification performed by finding hyperplane (decision boundary)
- Decision Boundary is given by $y(x) = 0$, where $y(x)$ is the model's prediction (discriminant function)
- $y(x_i) = w_0 + w_1 x_{i1} + w_2 x_{i2} = w_0 + w^T x_i$

In 2D-feature space, $y(x) = 0$ represents straight line

Cost function and Optimization:

$$J(w) = \sum_{i=1}^{N} (y_i - t_i)^2 \text{ ; with } w^* = \arg\min_w J(w)$$





$t \in \{0, 1\}$

$y(x) = \sigma(w_0 + w^T x)$

Logistic Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Bias trick:

$w \leftarrow [w_0, w]^T$ Absorb the bias

$x \leftarrow [1, x]^T$ Augment the input

$$y(x) = \sigma(w^T x) = \frac{1}{1 + e^{-w^T x}}$$